Spectral Shrunk Clustering: Reproduction, Numerical Audits, and Subset Uncertainty

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Abstract

Spectral Shrunk Clustering (SSC) refines a spectral representation through a convex fidelity–fusion objective. We separate paper reconstruction, numerical optimization and clustering quality. Joint ACC/NMI matching and declared source/subset sensitivities preserve distinct statistical and identity qualifications. Certified shrinkage has small, mixed net effects at gamma 1; normalization and graph interventions expose different limitations. An oracle completion gate passes on ORL, while label-free four-benchmark improvement remains unestablished. Source and implementation audits retain failures and adapted methods explicitly. Numerical agreement does not identify the authors' historical datasets.

1 What SSC optimizes

Spectral clustering (SC) represents samples using graph-Laplacian eigenvectors and then applies K-means. Chang et al. [Chang et al., 2015] argue that directly clustering the initial spectral coordinates can be unreliable, and propose learning *shrunk patterns* that preserve their anchors while contracting neighboring representations. Their contribution is a convex representation-refinement formulation, relaxation of an orthogonality constraint in this stage, and an iterative algorithm with clustering experiments. SSC here means Spectral Shrunk Clustering, rather than Sparse Subspace Clustering.

The input-space affinity A determines $L_A = I - D_A^{-1/2} A D_A^{-1/2}$. The c smallest eigenvectors form $F \in \mathbb{R}^{n\times c}$. A second affinity W is constructed from F and controls refinement (Figure 1). With each undirected edge stored once in incidence matrix B, Equation 8 is

$$P(G) = \sum_i \|g_i - f_i\|_2 + 2\gamma \sum_e w_e \|(BG)_e\|_2. \tag{1}$$

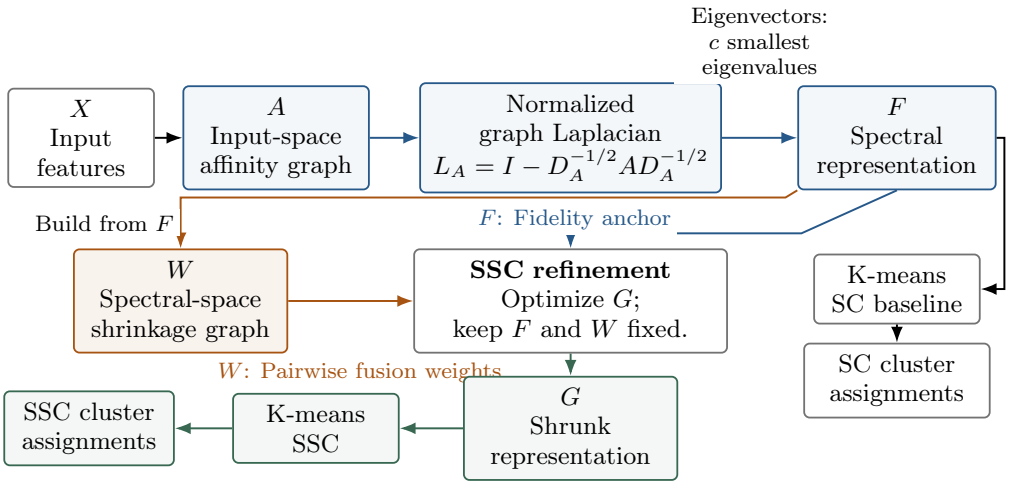
The factor 2 implements the paper's ordered-pair sum. Its stationary IRLS system uses $S + 2\gamma\widetilde{L}$ (E8), whereas printed Equation 13 uses $S + \gamma\widetilde{L}$ (E13). Thus E13 at nominal gamma targets E8 at effective gamma/2. We retain both tracks, rather than calling unequal targets a solver discrepancy.

For fixed F, W, nonnegative weights make (1) convex; unsquared fidelity does not guarantee uniqueness. Graph construction, eigenspace selection and final K-means are separate operations. A globally optimal G therefore need not improve ACC or NMI.

2 Published results and the reproduction protocol

We verified all 288 reported means and SD values in the paper's Tables 1–2: nine methods, eight benchmarks and two metrics. The paper states $k = 5$, 50 random initializations, a five-value parameter grid and native-objective selection. It does not fully resolve the SD hierarchy, source indices, Gaussian

Preprint.



SSC refinement objective (Equation 8)

$$\min_G \Big[\underbrace{\sum_i \|g_i - f_i\|_2}_{\text{Fidelity to } F} + \underbrace{2\gamma \sum_{i<j} W_{ij}\|g_i - g_j\|_2}_{\text{Fusion controlled by } W}\Big]$$

Each undirected edge once; both norms unsquared.

Figure 1: The two graph roles are distinct. A determines the initial spectral geometry F; W, built in that geometry, controls fusion while F remains fixed. Matched SC clusters the same F, and SSC clusters G.

width or graph symmetrization. Our choices are union kNN, median positive directed-neighbor squared-distance width, deterministic eigensolver initialization, and explicit one-start Lloyd rounding. ACC uses optimal label assignment and NMI uses geometric entropy normalization.

The paper's SSC ACC gains do not imply superiority on every metric: YaleB SC/SSC have ACC 34.8/49.8 but NMI 55.6/46.8. We therefore minimize joint discrepancy in percentage points,

$$D_{\max} = \max(|\overline{\text{ACC}} - \text{ACC}_p|, |\overline{\text{NMI}} - \text{NMI}_p|), \qquad D_{\text{mean}} = \tfrac{1}{2}(d_{\text{ACC}} + d_{\text{NMI}}).$$

We rank by $D_{\max}$, then D_{mean}, with a 10^{-12} tolerance and frozen compatibility tie breaks. Published-grid gamma values are not penalized for differing from 1. Both metrics and SD come from one configuration and its complete 50 seeds. The single native-best-of-50 assignment has no across-run SD; existing independent batches are a separate repeatability endpoint. Raw LDA inertia across learned subspaces is not treated as a common selection objective.

Data	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$	Fit
ORL	76.00 ± 1.60	74.63 ± 0.64	88.60 ± 1.50	89.71 ± 0.23	1.37	I
JAFFE	81.20 ± 1.60	74.75 ± 8.37	84.30 ± 1.60	85.60 ± 4.32	6.45	T
YaleB	49.80 ± 1.60	42.64 ± 0.61	46.80 ± 1.30	52.19 ± 0.31	7.16	T
USPS	75.50 ± 1.90	73.87 ± 7.53	79.80 ± 1.60	82.84 ± 3.96	3.04	I

Table 1: Historical SSC scalar records, reselected jointly by $D_{\max}$ then D_{mean}; this is a retrospective match, not label-free selection. Means $\pm$ sample SD use one configuration and all 50 seeds. I: iterate stopping without a gap certificate; T: iteration cap. None of these historical G arrays is recoverable, so current solver audits cannot certify their exact outputs. Settings and source versions appear in the appendix.

2.1 Dataset-subset uncertainty and retrospective numerical matching

Original sample indices were unavailable. Final jointly selected source/subset results and locked-selection rounding are reserved for the final subset release; no partial subset winner is reported here.

Such matching is retrospective, not label-free selection or an unbiased superiority test. Exact manifests support replay, not historical identity. Changed membership rebuilds A, F, W, G; source, resize, normalization and eligibility pools remain distinct (Appendix C).

Final SSC source/subset matching and locked-winner confirmation are reserved for the completed, validated scalar release. This edition presents the method, historical reconstruction and completed baseline, numerical and graph studies. No partial subset winner, performance average or confirmation result is reported.

3 What discrepancies can be explained

Numerics. For $c_e = 2\gamma w_e$, feasible q with $\|q_e\| \leq c_e$ and $\|(B^T q)_i\| \leq 1$ gives lower bound $D(q) = \langle B^T q, F \rangle$. We retain the initial feasible bound in preconditioned PDHG [Chambolle and Pock, 2011]; certification requires relative gap at most 10^{-5} and capacity violations at most 10^{-8}. On one frozen current USPS F/W, the gamma-1 continuation certifies its saved primal at gap 9.9996×10^{-6}, yet ACC/NMI change by only $-0.033/-0.086$ points against same-F SC. Gamma 1000 remains uncertified at gap 0.342; its collapsed saved point does not prove optimal collapse. Failed IRLS/dual-flow attempts and missing historical arrays remain separate limitations.

Implementation and sources. The pinned ClustPy implementation of LDA-Kmeans [Ding and Li, 2007] failed every LDA update in all 50 ORL and JAFFE fits; these are implementation-specific PCA–K-means degeneracies. The projection actually clustered and the source-returned projection are recorded separately. Existing-project LDA limits 100/300 give identical per-seed ACC/NMI on all five tested sources. The separately named ridge LDA-Kmeans→SC adaptation improves both means over raw/PCA$_{p_0}$ SC only on YaleB; it does not recover historical LDA-SC. Paired SLPS [Hou et al., 2013] $k = 5/7$ effects change with source and alpha. Native 16-by-16 EBSA MSRA retains all 1,799 rows and duplicates. Exact pins, adaptation formulas and controls are in Appendix C.

The baseline comparison contains 98 complete 50-seed SLPS configurations, 30 LDA configurations/control endpoints and 62 other configurations. The appendix retains numerical failures, iteration-limit outcomes and update diagnostics. Native 16-by-16 MSRA and ridge LDA-to-SC are explicitly named adaptations; exact historical LDA-SC remains unresolved.

4 Graph diagnostics and conditional extension

G1 compares SC/SSC on the same fixed representation. Certified gamma-1 shrinkage changes mean ACC/NMI by $+0.155/+0.014$ points on ORL and $-0.270/-0.276$ on YaleB, despite active movement. A and W have different roles: cross-class edges carry 4.6% of ORL W mass but 70.2% of YaleB W mass. Class-induced components, including isolates, distinguish within-class fragmentation from whole-graph connectivity. YaleB's W edge-length AUROC is 0.380, weakening the general premise that erroneous edges are longer; these associations alone do not establish their causal effect.

G2 separates anchor normalization from the fusion graph at matched mass. Row-normalized U alone raises SC ACC by 1.850/4.016 points on ORL/YaleB; angular-graph shrinkage at gamma 1 adds only 0.170/0.314 above U-SC. A stronger representation baseline therefore matters more than a comparison only with F_0-SC.

In the fixed G3 oracle intervention (15% additions, gamma 1000), ORL passes the diagnostic criterion: completion gains 10.415 ACC points over deletion and 24.863 over fixed oracle deletion plus mean blind additions, with nondecreasing NMI and qualified conditional intervals. YaleB's large raw gains remain numerically unqualified. The EnSC graph adaptation [You et al., 2016] is evaluated separately: oracle success does not establish an unsupervised-method gain. Native EnSC clustering and matched-movement controls distinguish affinity quality from the additional effect of SSC fusion.

The fixed oracle graph-intervention criterion is passed. The capability and locked-operation macro targets are unavailable because all four required comparisons must be qualified. Oracle graph evidence, unsupervised affinity construction and net shrinkage gains are distinct conclusions.

The primary EnSC–SSC capability means are 78.98/96.41/27.66/65.60 ACC on ORL/JAFFE/YaleB/USPS. The first three select gamma 1, but USPS selects gamma 0.001 with $G = F$: this is damage avoidance, not useful shrinkage. USPS at the fixed gamma 1 falls to

52.27 ACC and remains uncertified. Native EnSC has higher ACC/NMI than the primary adaptation on the first three datasets; the USPS primary exceeds native EnSC without moving from F_0.

The target is at least 2 points equal-weight ACC gain over the stronger fair SC/SSC baseline, with no mean NMI loss against either. Fair SC includes both F_0 and U. All four datasets have complete 50-start records, but required uncertified controls leave both capability and locked-operation macro targets unavailable. The parameter-selection-gap flag is therefore undefined. Additional seeds assess conditional rounding stability, not unseen-data generalization.

5 Conclusion

SSC gives a clear convex refinement target, while clustering quality depends on representation, graph, solver and rounding choices. Our contribution is reproducible source/subset sensitivity and explicit numerical and implementation audits, including negative or unresolved evidence. Exact historical LDA-SC and several dataset identities remain unresolved. Code, statistical records and reproducible commands are available in the public repository.

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A Historical measurements and endpoint separation

The original scalar measurements and ACC-only selections remain distinct historical endpoints. Joint selections use complete seed groups and preserve their original measurements. Missing historical representations limit optimization re-audits on their exact arrays. The historical 72-cell identity/status matrix remains separate from newly measured candidates and adaptations. The original ACC-only closest-display and fixed-reference tables remain available in the committed historical release. The current joint 72-cell table below supersedes their display selection without rewriting any trial.

B Results, statistical endpoints and coverage

B.1 Jointly reselected historical mean endpoint

Table 2 reports the complete 72-cell inventory. Values are percent. Every measured pair uses the same configuration and 50 batch-zero seeds; $\pm$ denotes sample SD. Missing cells retain their identity or method restriction. Historical scalar evidence is separate from new recoverable artifacts. The former ACC-only and fixed-reference tables remain in the committed historical release.

Table 2: Joint closest historical mean/sample-SD endpoint; unavailable cells are not filled with adapted measurements.

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR	K-means	36.30 ± 1.40	–	68.70 ± 3.00	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SC	41.60 ± 2.10	–	71.30 ± 2.60	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	PCA–K-means	39.80 ± 1.80	–	69.40 ± 2.80	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	PCA–SC	43.20 ± 1.70	–	70.30 ± 2.40	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LDA–K-means	40.40 ± 1.50	–	69.90 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LDA–SC	44.50 ± 1.30	–	70.80 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LLC	48.70 ± 1.60	–	71.20 ± 2.40	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SLPS	49.20 ± 1.40	–	72.40 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SSC	51.30 ± 1.50	–	73.20 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
JAFFE	K-means	75.60 ± 1.80	71.97 ± 7.55	79.40 ± 0.80	81.97 ± 3.70	3.63
JAFFE; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 3.10$.						
JAFFE	SC	76.10 ± 1.60	68.53 ± 11.15	80.20 ± 0.90	83.02 ± 5.72	7.57
JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 5.20$.						
JAFFE	PCA–K-means	75.80 ± 1.50	73.20 ± 5.58	79.80 ± 0.80	82.95 ± 3.67	3.15
JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.88$.						
JAFFE	PCA–SC	76.90 ± 1.80	72.30 ± 0.00	81.50 ± 1.30	82.86 ± 0.00	4.60
JAFFE; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.98$.						
JAFFE	LDA–K-means	76.50 ± 1.70	64.44 ± 9.19	82.10 ± 1.40	78.60 ± 5.08	12.06
JAFFE; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 7.78$.						
JAFFE	LDA–SC	77.40 ± 1.90	–	81.90 ± 0.90	–	–
No eligible complete measured configuration; historical restrictions remain						
JAFFE	LLC	78.60 ± 1.50	91.55 ± 0.00	82.50 ± 1.70	92.75 ± 0.00	12.95
JAFFE; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 11.60$.						
JAFFE	SLPS	79.50 ± 2.10	80.86 ± 5.77	83.10 ± 2.10	86.66 ± 3.34	3.56
JAFFE; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.46$.						
JAFFE	SSC	81.20 ± 1.60	74.75 ± 8.37	84.30 ± 1.60	85.60 ± 4.32	6.45
JAFFE; Y0, E13, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $D_{\text{mean}} = 3.87$.						
ORL	K-means	60.50 ± 1.80	61.46 ± 2.85	80.30 ± 1.80	82.76 ± 1.17	2.46
ORL_att92; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 1.71$.						
ORL	SC	72.70 ± 2.30	72.73 ± 3.15	85.80 ± 1.90	87.49 ± 1.22	1.69
ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 0.86$.						
ORL	PCA–K-means	64.50 ± 2.30	63.08 ± 2.26	80.60 ± 1.60	82.34 ± 0.90	1.74
ORL_att92; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 1.58$.						

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
ORL	PCA–SC	67.80 ± 2.10	67.57 ± 2.92	86.30 ± 1.40	85.81 ± 1.27	0.49
ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.36$.						
ORL	LDA–K-means	65.60 ± 2.60	56.31 ± 4.14	81.10 ± 2.10	81.00 ± 1.65	9.29
ORL_att92; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\mathrm{mean}} = 4.70$.						
ORL	LDA–SC	68.30 ± 2.40	–	86.80 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
ORL	LLC	71.50 ± 2.20	81.07 ± 0.61	84.90 ± 1.50	89.92 ± 0.27	9.57
ORL_att92; Y0, YU_SHI; NOT_APPLICABLE; $D_{\mathrm{mean}} = 7.29$.						
ORL	SLPS	74.20 ± 1.80	73.66 ± 2.13	87.20 ± 1.80	87.71 ± 0.91	0.54
ORL_att92; Y0, one_shot, G; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.53$.						
ORL	SSC	76.00 ± 1.60	74.63 ± 0.64	88.60 ± 1.50	89.71 ± 0.23	1.37
ORL_att92; Y0, E8, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $D_{\mathrm{mean}} = 1.24$.						
UMIST	K-means	41.30 ± 1.60	–	64.40 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SC	52.20 ± 1.40	–	72.10 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	PCA–K-means	48.80 ± 1.70	–	68.20 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	PCA–SC	54.10 ± 1.90	–	72.90 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LDA–K-means	49.70 ± 1.80	–	68.80 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LDA–SC	54.70 ± 1.50	–	74.10 ± 2.00	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LLC	63.30 ± 1.80	–	77.30 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SLPS	70.40 ± 1.60	–	82.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SSC	71.10 ± 1.80	–	84.10 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	K-means	41.70 ± 1.10	–	58.60 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SC	43.60 ± 1.50	–	59.70 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	PCA–K-means	42.40 ± 1.50	–	59.10 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	PCA–SC	44.30 ± 1.80	–	60.60 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LDA–K-means	42.90 ± 1.70	–	59.80 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LDA–SC	45.10 ± 1.40	–	61.30 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LLC	40.70 ± 1.80	–	61.40 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SLPS	48.50 ± 1.90	–	63.60 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SSC	49.40 ± 1.30	–	64.10 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	K-means	46.20 ± 1.70	–	56.70 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SC	52.30 ± 1.80	–	70.00 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	PCA–K-means	56.00 ± 1.90	–	60.30 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	PCA–SC	54.90 ± 1.60	–	72.40 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LDA–K-means	56.40 ± 1.80	–	61.10 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LDA–SC	55.10 ± 1.70	–	73.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LLC	48.10 ± 1.40	–	66.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SLPS	60.30 ± 1.30	–	69.60 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SSC	63.20 ± 1.10	–	72.20 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
YaleB	K-means	14.40 ± 1.50	9.74 ± 0.46	17.30 ± 1.50	13.25 ± 0.59	4.66
YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.35$.						
YaleB	SC	34.80 ± 1.40	37.39 ± 2.22	55.60 ± 1.60	52.96 ± 1.00	2.64
YaleB_dengcai32; Y3, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.61$.						
YaleB	PCA–K-means	24.90 ± 1.80	9.85 ± 0.55	26.70 ± 1.80	13.32 ± 0.66	15.05
YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 14.22$.						
YaleB	PCA–SC	36.80 ± 2.00	30.92 ± 0.54	38.60 ± 1.50	38.86 ± 0.95	5.88
YaleB_dengcai32; Y1, G; NOT_APPLICABLE; $D_{\text{mean}} = 3.07$.						
YaleB	LDA–K-means	26.10 ± 1.80	9.80 ± 0.85	29.40 ± 1.60	13.79 ± 1.15	16.30
YaleB_dengcai32; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 15.96$.						
YaleB	LDA–SC	38.20 ± 1.60	–	39.90 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
YaleB	LLC	38.20 ± 1.50	32.35 ± 0.05	34.10 ± 1.30	39.68 ± 0.11	5.85
YaleB_dengcai32; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 5.72$.						
YaleB	SLPS	47.30 ± 1.70	13.52 ± 0.78	41.40 ± 1.60	19.44 ± 0.69	33.78
YaleB_dengcai32; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 27.87$.						
YaleB	SSC	49.80 ± 1.60	42.64 ± 0.61	46.80 ± 1.30	52.19 ± 0.31	7.16
YaleB_dengcai32; Y3, E8, $\gamma = 1e+06$, G; MAX_ITER_REACHED; $D_{\text{mean}} = 6.27$.						
USPS	K-means	65.40 ± 1.70	66.06 ± 2.17	67.30 ± 1.80	61.04 ± 0.87	6.26
USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 3.46$.						
USPS	SC	64.30 ± 1.40	61.74 ± 8.09	69.50 ± 1.60	74.94 ± 4.49	5.44
USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 4.00$.						
USPS	PCA–K-means	69.40 ± 1.80	65.25 ± 2.85	73.10 ± 1.90	61.10 ± 0.95	12.00
USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 8.08$.						
USPS	PCA–SC	69.10 ± 1.50	65.53 ± 3.26	74.20 ± 1.80	78.92 ± 2.12	4.72
USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.15$.						
USPS	LDA–K-means	70.10 ± 1.30	58.61 ± 6.33	75.10 ± 1.60	60.29 ± 4.21	14.81
USPS; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 13.15$.						
USPS	LDA–SC	70.40 ± 1.50	–	75.40 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
USPS	LLC	63.90 ± 1.70	65.27 ± 0.01	67.50 ± 1.50	72.76 ± 0.01	5.26
USPS; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 3.32$.						
USPS	SLPS	71.40 ± 1.60	70.81 ± 8.33	76.50 ± 1.80	76.30 ± 2.47	0.59
USPS; Y0, one_shot, R; NOT_APPLICABLE; $D_{\text{mean}} = 0.39$.						
USPS	SSC	75.50 ± 1.90	73.87 ± 7.53	79.80 ± 1.60	82.84 ± 3.96	3.04
USPS; Y0, E8, $\gamma = 1000$, R; ITERATE_CONVERGED_UNCERTIFIED; $D_{\text{mean}} = 2.34$.						

B.2 Native-objective-selected endpoint

Within one fixed G, native K-means SSE selects one of50 assignments. This is a single assignment with no across-run SD. It does not compete with the mean endpoint. Full method-specific endpoint records are in `results/submission_refresh/endpoint_comparison.csv`; LDA cross-subspace native selection is unavailable.

SSC data	ACC	NMI	$D_{\max}$	D_{mean}	Seed	Fit
AR	–	–	–	–	–	–
JAFFE	95.31	94.09	14.11	11.95	13	T
ORL	75.00	88.60	1.00	0.50	40	I
UMIST	–	–	–	–	–	–
BinAlpha	–	–	–	–	–	–
MSRA50	–	–	–	–	–	–
YaleB	43.33	54.15	7.35	6.91	45	I
USPS	79.62	85.38	5.58	4.85	38	I

B.3 Separate historical repeatability batches

Existing multi-batch records are a separate endpoint. Each batch contains50 disjoint rounding seeds on its fixed input. The following ranges describe variation of batch means and native-selected assignments, without treating a single native assignment as having a within-batch SD. They are not independent datasets, and are not new campaign measurements. All methods and exact configuration IDs are preserved in `historical_repeatability_batches.csv` and its summary/manifest. LDA source-native selection remains diagnostic only.

SSC setting	Batches	Batch ACC	Batch NMI	Native ACC	Native NMI
BinAlpha_candidate26x30 / E13 / gamma=1e+00 / R	10	42.21–43.23	56.50–57.12	44.62–47.82	57.53–59.66
BinAlpha_candidate26x30 / E13 / gamma=1e+00 / G	10	47.05–47.52	58.55–58.91	45.51–49.36	58.37–59.70
BinAlpha_candidate26x30 / E8 / gamma=1e+00 / R	10	40.54–41.55	55.87–56.61	43.59–45.90	57.04–59.18
BinAlpha_candidate26x30 / E8 / gamma=1e+00 / G	10	46.35–47.18	58.33–59.09	45.38–48.33	57.06–59.31
JAFFE / E13 / gamma=1e+00 / R	10	61.13–68.59	78.60–83.04	80.75–95.31	89.37–94.16
JAFFE / E13 / gamma=1e+00 / G	10	95.06–95.31	94.05–94.14	95.31–95.31	94.12–94.12
JAFFE / E8 / gamma=1e+00 / R	10	61.08–68.43	78.70–83.05	80.75–95.31	89.37–94.16
JAFFE / E8 / gamma=1e+00 / G	10	94.85–95.31	93.99–94.14	95.31–95.31	94.12–94.12
ORL_att92 / E13 / gamma=1e+00 / R	10	55.83–57.48	80.42–81.43	63.00–73.25	82.97–87.94
ORL_att92 / E13 / gamma=1e+00 / G	10	78.04–78.77	89.52–89.86	78.25–80.50	89.83–90.93
ORL_att92 / E8 / gamma=1e+00 / R	10	56.12–58.05	80.70–81.68	62.75–70.50	83.63–87.91
ORL_att92 / E8 / gamma=1e+00 / G	10	78.25–78.75	89.66–89.95	78.25–80.50	89.97–91.39
UMIST_precropped575 / E13 / gamma=1e+00 / R	10	54.34–57.24	75.69–77.00	56.70–67.48	76.40–82.39
UMIST_precropped575 / E13 / gamma=1e+00 / G	10	59.98–61.01	78.21–78.68	60.70–60.87	77.83–78.02
UMIST_precropped575 / E8 / gamma=1e+00 / R	10	54.49–56.87	74.95–76.16	54.61–66.61	76.62–81.01
UMIST_precropped575 / E8 / gamma=1e+00 / G	10	60.35–61.78	78.35–78.90	60.35–61.04	77.92–78.52
USPS / E13 / gamma=1e+00 / R	10	59.71–62.17	74.34–75.71	65.47–65.59	80.97–81.19
USPS / E13 / gamma=1e+00 / G	10	64.89–66.26	79.94–80.86	65.48–65.50	80.99–81.01
USPS / E8 / gamma=1e+00 / R	10	59.56–61.93	74.38–75.53	65.63–65.63	81.17–81.24
USPS / E8 / gamma=1e+00 / G	10	64.76–65.94	79.77–80.89	65.62–65.63	81.17–81.19
YaleB_cropped2414 / E13 / gamma=1e+00 / R	10	17.96–18.20	26.41–26.67	19.51–21.29	26.78–27.84
YaleB_cropped2414 / E13 / gamma=1e+00 / G	10	21.59–21.83	27.49–27.75	21.71–21.96	27.41–28.23
YaleB_cropped2414 / E8 / gamma=1e+00 / R	10	17.46–17.89	26.05–26.34	18.77–20.34	25.94–28.07
YaleB_cropped2414 / E8 / gamma=1e+00 / G	10	21.40–21.56	27.24–27.58	21.25–22.08	27.07–28.19
YaleB_dengcai32 / E13 / gamma=1e+00 / R	10	19.80–20.29	29.21–29.63	21.46–23.28	28.80–31.77
YaleB_dengcai32 / E13 / gamma=1e+00 / G	10	25.32–25.55	31.13–31.47	24.98–25.52	31.31–31.96
YaleB_dengcai32 / E8 / gamma=1e+00 / R	10	19.50–19.88	29.09–29.32	20.59–23.16	29.52–30.98
YaleB_dengcai32 / E8 / gamma=1e+00 / G	10	25.31–25.46	31.24–31.43	25.02–25.10	31.54–31.65

B.4 Dataset-subset matching: both endpoints and eligibility pools

Final SSC source/subset matching and locked-winner confirmation are reserved for the completed, validated scalar release. This edition presents the method, historical reconstruction and completed baseline, numerical and graph studies. No partial subset winner, performance average or confirmation result is reported.

B.5 Additional rounding on locked winners

Final locked-winner confirmation is reserved; no stability result is reported.

B.6 Baseline comparisons

Data	Method	Setting	ACC	NMI	Status
JAFFE	Ridge LDA (adapted)	limit 300	83.85 ± 2.50	87.77 ± 1.53	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	Ridge LDA–SC (adapted)	adapted_sc	59.38 ± 9.77	73.72 ± 5.60	COMPLETED
JAFFE	PCA_{p_0}–SC	pca_p0_sc	69.00 ± 11.08	83.02 ± 5.86	COMPLETED
JAFFE	PCA_r–SC	pca_r_sc	58.64 ± 7.81	73.20 ± 5.04	COMPLETED
JAFFE	Raw SC	raw_sc	68.53 ± 11.15	83.02 ± 5.72	COMPLETED
MSRA25_EBSA_16x16	Ridge LDA (adapted)	limit 300	57.25 ± 4.00	72.24 ± 2.52	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	Ridge LDA–SC (adapted)	adapted_sc	43.61 ± 4.77	54.73 ± 3.91	COMPLETED
MSRA25_EBSA_16x16	PCA_{p_0}–SC	pca_p0_sc	49.54 ± 5.74	62.89 ± 4.11	COMPLETED
MSRA25_EBSA_16x16	PCA_r–SC	pca_r_sc	37.81 ± 3.64	49.97 ± 2.60	COMPLETED
MSRA25_EBSA_16x16	Raw SC	raw_sc	49.14 ± 6.24	62.84 ± 4.72	COMPLETED
ORL_att92	Ridge LDA (adapted)	limit 300	75.37 ± 3.38	88.94 ± 1.53	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	Ridge LDA–SC (adapted)	adapted_sc	49.19 ± 3.76	75.66 ± 2.65	COMPLETED
ORL_att92	PCA_{p_0}–SC	pca_p0_sc	55.44 ± 4.28	80.12 ± 2.25	COMPLETED
ORL_att92	PCA_r–SC	pca_r_sc	55.29 ± 4.36	80.46 ± 2.21	COMPLETED
ORL_att92	Raw SC	raw_sc	56.18 ± 4.19	80.47 ± 2.29	COMPLETED
USPS	Ridge LDA (adapted)	limit 300	70.17 ± 0.11	68.70 ± 0.09	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
USPS	Ridge LDA–SC (adapted)	adapted_sc	57.79 ± 6.19	61.10 ± 2.66	COMPLETED
USPS	PCA_{p_0}–SC	pca_p0_sc	62.78 ± 7.95	75.52 ± 4.48	COMPLETED
USPS	PCA_r–SC	pca_r_sc	55.93 ± 4.32	65.98 ± 2.83	COMPLETED
USPS	Raw SC	raw_sc	61.74 ± 8.09	74.94 ± 4.49	COMPLETED
YaleB_cropped2414	Ridge LDA (adapted)	limit 300	42.12 ± 1.62	51.32 ± 1.22	COMPLETED
Stops: MAX ITER (32), PARTITION STABLE (18); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	Ridge LDA–SC (adapted)	adapted_sc	35.85 ± 2.62	50.58 ± 1.59	COMPLETED
YaleB_cropped2414	PCA_{p_0}–SC	pca_p0_sc	17.74 ± 1.15	26.37 ± 0.64	COMPLETED
YaleB_cropped2414	PCA_r–SC	pca_r_sc	15.71 ± 1.06	22.62 ± 0.87	COMPLETED
YaleB_cropped2414	Raw SC	raw_sc	18.34 ± 1.28	26.90 ± 0.67	COMPLETED
JAFFE	LDA existing	limit 100	64.44 ± 9.19	78.60 ± 5.08	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	LDA existing	limit 30	64.44 ± 9.19	78.60 ± 5.08	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	LDA existing	limit 300	64.44 ± 9.19	78.60 ± 5.08	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LDA existing	limit 100	46.88 ± 5.32	60.43 ± 4.28	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LDA existing	limit 30	46.87 ± 5.33	60.41 ± 4.30	COMPLETED
Stops: LABELS STABLE (49), MAX ITER (1); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LDA existing	limit 300	46.88 ± 5.32	60.43 ± 4.28	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	LDA existing	limit 100	56.31 ± 4.14	81.00 ± 1.65	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	LDA existing	limit 30	56.31 ± 4.14	81.00 ± 1.65	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	LDA existing	limit 300	56.31 ± 4.14	81.00 ± 1.65	COMPLETED

Data	Method	Setting	ACC	NMI	Status
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
USPS	LDA existing	limit 100	58.29 ± 6.61	59.93 ± 4.52	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
USPS	LDA existing	limit 30	58.61 ± 6.33	60.29 ± 4.21	COMPLETED
Stops: LABELS STABLE (12), MAX ITER (38); failed updates: 0; zero-success runs: 0.					
USPS	LDA existing	limit 300	58.29 ± 6.61	59.93 ± 4.52	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	LDA existing	limit 100	9.67 ± 0.82	13.42 ± 1.06	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	LDA existing	limit 300	9.67 ± 0.82	13.42 ± 1.06	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	LDA existing	limit 30	9.67 ± 0.82	13.42 ± 1.06	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	ClustPy reference	limit 300	82.95 ± 1.07	87.83 ± 0.83	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 625; zero-success runs: 50.					
MSRA25_EBSA_16x16	ClustPy reference	limit 300	54.19 ± 3.44	65.42 ± 2.47	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	ClustPy reference	limit 300	74.22 ± 2.58	87.76 ± 1.11	COMPLETED
Stops: MAX ITER (50); failed updates: 15000; zero-success runs: 50.					
USPS	ClustPy reference	limit 300	67.23 ± 0.14	65.77 ± 0.13	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	ClustPy reference	limit 300	10.57 ± 0.75	14.84 ± 1.24	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LLC	0.0625	51.53 ± 0.40	66.90 ± 0.43	COMPLETED
MSRA25_EBSA_16x16	LLC	0.25	55.50 ± 1.50	66.52 ± 0.92	COMPLETED
MSRA25_EBSA_16x16	LLC	1.0	58.91 ± 2.11	68.28 ± 1.07	COMPLETED
MSRA25_EBSA_16x16	LLC	16.0	58.72 ± 2.74	69.12 ± 1.55	COMPLETED
MSRA25_EBSA_16x16	LLC	4.0	59.16 ± 2.37	69.04 ± 1.00	COMPLETED
MSRA25_EBSA_16x16	LLC	0.0625	56.01 ± 2.67	66.20 ± 1.22	COMPLETED
MSRA25_EBSA_16x16	LLC	0.25	51.80 ± 3.26	58.84 ± 1.56	COMPLETED
MSRA25_EBSA_16x16	LLC	1.0	53.73 ± 2.57	63.53 ± 1.83	COMPLETED
MSRA25_EBSA_16x16	LLC	16.0	52.15 ± 3.09	62.69 ± 1.98	COMPLETED
MSRA25_EBSA_16x16	LLC	4.0	52.52 ± 2.73	62.92 ± 1.96	COMPLETED
MSRA25_EBSA_16x16	LLC	0.0625	56.42 ± 2.20	64.75 ± 1.23	COMPLETED
MSRA25_EBSA_16x16	LLC	0.25	52.82 ± 2.55	62.63 ± 1.58	COMPLETED
MSRA25_EBSA_16x16	LLC	1.0	52.88 ± 2.63	63.12 ± 1.85	COMPLETED
MSRA25_EBSA_16x16	LLC	16.0	51.86 ± 3.05	62.73 ± 1.55	COMPLETED
MSRA25_EBSA_16x16	LLC	4.0	53.26 ± 2.84	63.28 ± 1.72	COMPLETED
MSRA25_EBSA_16x16	K-means	kmeans	51.23 ± 4.85	59.81 ± 3.41	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d10	49.68 ± 4.36	56.62 ± 2.86	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d100	51.22 ± 4.92	59.77 ± 3.50	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d110	51.16 ± 4.91	59.70 ± 3.49	COMPLETED

Data	Method	Setting	ACC	NMI	Status
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d120	51.17 ± 4.90	59.71 ± 3.47	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d130	51.07 ± 4.83	59.69 ± 3.40	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d140	51.04 ± 4.87	59.62 ± 3.46	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d150	51.08 ± 4.84	59.65 ± 3.39	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d20	50.87 ± 5.31	58.73 ± 3.75	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d30	51.19 ± 4.93	59.57 ± 3.62	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d40	51.43 ± 4.51	59.68 ± 3.37	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d50	51.27 ± 4.73	59.69 ± 3.45	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d60	51.12 ± 4.74	59.50 ± 3.44	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d70	51.28 ± 4.89	59.74 ± 3.41	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d80	51.21 ± 4.91	59.76 ± 3.53	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d90	51.22 ± 4.87	59.70 ± 3.51	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d10	34.55 ± 2.85	45.37 ± 2.60	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d100	49.54 ± 5.74	62.89 ± 4.11	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d110	48.93 ± 5.94	63.99 ± 5.10	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d120	48.42 ± 4.57	62.33 ± 3.33	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d130	45.67 ± 4.62	60.11 ± 3.56	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d140	46.56 ± 5.48	59.72 ± 4.17	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d150	45.73 ± 4.63	58.83 ± 3.40	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d20	43.63 ± 4.56	57.19 ± 3.12	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d30	46.00 ± 4.68	59.01 ± 3.38	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d40	45.56 ± 3.88	59.90 ± 2.84	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d50	48.51 ± 5.07	62.75 ± 3.42	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d60	46.86 ± 4.02	61.25 ± 2.84	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d70	47.86 ± 5.57	60.95 ± 5.14	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d80	49.23 ± 4.15	62.78 ± 3.67	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d90	44.87 ± 4.91	59.15 ± 3.93	COMPLETED
MSRA25_EBSA_16x16	K-means	kmeans	51.23 ± 4.85	59.81 ± 3.41	COMPLETED
JAFFE	SLPS	k5__alpha0.2__d110	72.21 ± 8.46	84.78 ± 4.19	COMPLETED
JAFFE	SLPS	k5__alpha0.2__d30	72.20 ± 8.46	84.78 ± 4.19	COMPLETED
JAFFE	SLPS	k5__alpha0.2__d70	72.21 ± 8.46	84.78 ± 4.19	COMPLETED
JAFFE	SLPS	k5__alpha0.5__d110	74.41 ± 9.72	85.63 ± 4.45	COMPLETED
JAFFE	SLPS	k5__alpha0.5__d30	74.45 ± 9.90	85.52 ± 4.58	COMPLETED

Data	Method	Setting	ACC	NMI	Status
JAFFE	SLPS	k5__alpha0.5__d70	74.41 ± 9.72	85.63 ± 4.45	COMPLETED
JAFFE	SLPS	k5__alpha0.8__d110	73.06 ± 7.18	84.07 ± 4.05	COMPLETED
JAFFE	SLPS	k5__alpha0.8__d30	73.13 ± 7.24	84.18 ± 4.02	COMPLETED
JAFFE	SLPS	k5__alpha0.8__d70	72.85 ± 6.89	83.84 ± 3.96	COMPLETED
JAFFE	SLPS	k5__alpha0.95__d70__SKLEARN_GREEDY_KMEANSPP	81.53 ± 6.24	86.97 ± 3.32	COMPLETED
JAFFE	SLPS	k7__alpha0.2__d110	73.33 ± 8.76	85.46 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.2__d30	73.11 ± 8.97	85.36 ± 4.24	COMPLETED
JAFFE	SLPS	k7__alpha0.2__d70	73.33 ± 8.76	85.46 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.5__d110	74.21 ± 8.75	85.71 ± 4.05	COMPLETED
JAFFE	SLPS	k7__alpha0.5__d30	74.01 ± 8.37	85.59 ± 3.83	COMPLETED
JAFFE	SLPS	k7__alpha0.5__d70	74.21 ± 8.75	85.71 ± 4.05	COMPLETED
JAFFE	SLPS	k7__alpha0.8__d110	73.92 ± 8.02	84.72 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.8__d30	74.58 ± 8.28	85.14 ± 4.32	COMPLETED
JAFFE	SLPS	k7__alpha0.8__d70	73.92 ± 8.02	84.72 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.95__d70__SKLEARN_GREEDY_KMEANSPP	80.85 ± 6.58	86.91 ± 3.58	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.2__d110	52.30 ± 5.18	67.22 ± 3.91	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.2__d30	52.62 ± 5.08	67.45 ± 3.85	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.2__d70	52.30 ± 5.17	67.19 ± 3.88	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.5__d110	52.10 ± 5.41	64.55 ± 4.36	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.5__d30	52.17 ± 5.23	64.46 ± 4.37	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.5__d70	52.16 ± 5.31	64.54 ± 4.27	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.8__d110	51.42 ± 5.00	61.63 ± 3.81	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.8__d30	51.18 ± 5.05	61.33 ± 3.83	COMPLETED
MSRA25_EBSA_16x16	SLPS	k5__alpha0.8__d70	51.43 ± 4.96	61.70 ± 3.77	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.2__d110	52.39 ± 5.28	67.77 ± 3.87	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.2__d30	52.48 ± 4.91	67.72 ± 3.72	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.2__d70	52.30 ± 5.15	67.69 ± 3.76	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.5__d110	52.10 ± 5.27	65.53 ± 4.35	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.5__d30	52.37 ± 5.40	65.75 ± 4.49	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.5__d70	52.08 ± 5.32	65.60 ± 4.36	COMPLETED

Data	Method	Setting	ACC	NMI	Status
MSRA25_EBSA_16x16	SLPS	k7__alpha0.8__d110	51.66 ± 5.06	62.58 ± 3.67	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.8__d30	51.71 ± 5.30	62.75 ± 3.88	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.8__d70	51.87 ± 5.05	62.79 ± 3.79	COMPLETED
ORL_att92	SLPS	k5__alpha0.2__d110	66.22 ± 3.56	87.15 ± 0.92	COMPLETED
ORL_att92	SLPS	k5__alpha0.2__d30	66.65 ± 3.44	87.26 ± 0.95	COMPLETED
ORL_att92	SLPS	k5__alpha0.2__d70	66.29 ± 3.62	87.21 ± 0.92	COMPLETED
ORL_att92	SLPS	k5__alpha0.5__d110	65.90 ± 3.91	86.84 ± 1.21	COMPLETED
ORL_att92	SLPS	k5__alpha0.5__d30	66.13 ± 3.77	86.91 ± 1.17	COMPLETED
ORL_att92	SLPS	k5__alpha0.5__d70	66.07 ± 3.88	86.92 ± 1.19	COMPLETED
ORL_att92	SLPS	k5__alpha0.85__d40__SKLEARN_GREEDY_KMEANSPP	74.68 ± 3.12	88.45 ± 1.19	COMPLETED
ORL_att92	SLPS	k5__alpha0.8__d110	64.91 ± 3.18	85.63 ± 1.06	COMPLETED
ORL_att92	SLPS	k5__alpha0.8__d30	66.01 ± 3.32	86.12 ± 1.05	COMPLETED
ORL_att92	SLPS	k5__alpha0.8__d70	65.20 ± 3.15	85.78 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.2__d110	64.53 ± 3.72	85.88 ± 1.09	COMPLETED
ORL_att92	SLPS	k7__alpha0.2__d30	64.44 ± 3.77	85.78 ± 1.08	COMPLETED
ORL_att92	SLPS	k7__alpha0.2__d70	64.45 ± 3.77	85.90 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.5__d110	65.19 ± 3.72	85.82 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.5__d30	65.51 ± 3.77	85.81 ± 1.11	COMPLETED
ORL_att92	SLPS	k7__alpha0.5__d70	65.26 ± 3.76	85.82 ± 1.08	COMPLETED
ORL_att92	SLPS	k7__alpha0.85__d40__SKLEARN_GREEDY_KMEANSPP	74.19 ± 1.98	87.90 ± 0.87	COMPLETED
ORL_att92	SLPS	k7__alpha0.8__d110	64.61 ± 3.01	85.38 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.8__d30	65.57 ± 3.40	85.58 ± 1.14	COMPLETED
ORL_att92	SLPS	k7__alpha0.8__d70	64.59 ± 2.78	85.40 ± 1.06	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d110	11.83 ± 1.07	18.36 ± 1.01	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d30	11.76 ± 1.02	18.28 ± 0.97	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d70	11.83 ± 1.06	18.37 ± 0.96	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	12.11 ± 0.66	18.46 ± 0.62	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.5__d110	10.98 ± 0.92	16.51 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.5__d30	10.83 ± 0.77	16.29 ± 0.75	COMPLETED

Data	Method	Setting	ACC	NMI	Status
YaleB_cropped2414	SLPS	k5__alpha0.5__d70	10.96 ± 0.87	16.45 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.8__d110	9.91 ± 0.75	14.04 ± 0.85	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.8__d30	9.74 ± 0.65	13.90 ± 0.82	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.8__d70	9.88 ± 0.70	14.01 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d110	11.13 ± 1.01	16.95 ± 0.94	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d30	11.11 ± 1.01	16.92 ± 0.93	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d70	11.13 ± 1.03	16.94 ± 0.93	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	11.24 ± 0.54	17.07 ± 0.44	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.5__d110	10.79 ± 0.89	16.02 ± 0.89	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.5__d30	10.70 ± 0.89	15.96 ± 0.87	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.5__d70	10.82 ± 0.86	16.05 ± 0.86	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.8__d110	10.09 ± 0.78	14.22 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.8__d30	9.94 ± 0.73	14.08 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.8__d70	10.10 ± 0.77	14.21 ± 0.81	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d110	13.07 ± 1.17	20.48 ± 1.11	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d30	12.90 ± 1.10	20.32 ± 1.06	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d70	13.05 ± 1.19	20.54 ± 1.13	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	14.42 ± 0.73	21.36 ± 0.72	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.5__d110	11.81 ± 0.95	18.17 ± 0.91	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.5__d30	11.51 ± 0.93	17.76 ± 0.82	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.5__d70	11.76 ± 0.93	18.07 ± 0.88	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.8__d110	10.51 ± 0.80	15.06 ± 0.79	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.8__d30	10.18 ± 0.80	14.70 ± 0.88	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.8__d70	10.40 ± 0.82	14.90 ± 0.73	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d110	12.36 ± 1.23	18.90 ± 1.11	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d30	12.24 ± 1.20	18.71 ± 1.11	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d70	12.33 ± 1.23	18.83 ± 1.08	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	13.52 ± 0.78	19.44 ± 0.69	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.5__d110	11.85 ± 1.02	17.74 ± 0.90	COMPLETED

Data	Method	Setting	ACC	NMI	Status
YaleB_dengcai32	SLPS	k7__alpha0.5__d30	11.61 ± 0.99	17.51 ± 0.89	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.5__d70	11.87 ± 0.99	17.79 ± 0.89	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.8__d110	10.65 ± 0.79	15.35 ± 0.75	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.8__d30	10.48 ± 0.79	15.15 ± 0.86	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.8__d70	10.66 ± 0.83	15.36 ± 0.74	COMPLETED

B.7 USPS numerical audit on one newly frozen F/W pair

These rows use a newly generated, recoverable current USPS representation and its exact shrinkage graph. They do not identify or replace the missing historical selected F/G arrays. Every non-SC point targets unsquared E8 with undirected coefficient $2\gamma w$. All original ten stages remain visible; the two additional restarts are separate numerical attempts, not replacements for the historical finite-iteration winner or subset-search output.

The numerical audit and subset USPS representation record exactly matching current F/W content addresses. Recorded input-reference status: MATCHING_RECORDED_CURRENT_F_W_OBJECTS. This is only a comparison of current input references. Each certificate belongs to its own saved continuation G; it does not certify the locked IRLS G, establish a certified closest search result, or recover missing historical F/G arrays.

Tables 6 and 7 refer to the same stage/gamma rows. C/U denote the recorded certified/uncertified state, with a required relative gap at most 10^{-5} and capacity violation at most 10^{-8} for certification. A solver stopping status alone is not a certificate. Uncertified points cannot establish that the mathematical objective lacks clustering capacity. Missing exports or saved points stay unavailable.

Table 6: Same-target numerical evidence. Original rows show fit CPU only where recorded; extra-restart CPU includes the full numerical attempt. A dash means unavailable metadata, not zero cost.

Stage	γ	Primal	Dual	Rel. gap	Cert.	CPU (s)	Solver status
SC on F	–	–	–	–	–	–	NOT_APPLICABLE
IRLS300	1	13.577	–	–	U	1800.0	CPU_LIMIT
Uniform	1	13.644	13.562	0.006	U	160.0	MAX_ITER_REACHED
Diagonal	1	13.587	13.576	0.000864	U	160.5	MAX_ITER_REACHED
Flow	1	267.96	1.6424e-16	1	U	12.7	INCOMPLETE_LINEAR_SOLVE
Warm restart	1	13.577	13.577	2.58e-05	U	155.2	MAX_ITER_REACHED
IRLS300	1000	388.48	–	–	U	11.7	FAILED
Uniform	1000	267.96	87.328	0.674	U	154.2	MAX_ITER_REACHED
Diagonal	1000	267.96	86.357	0.678	U	142.7	MAX_ITER_REACHED
Flow	1000	267.96	1.6424e-13	1	U	11.3	INCOMPLETE_LINEAR_SOLVE
Warm restart	1000	267.96	120.26	0.551	U	136.5	MAX_ITER_REACHED
Extra restart	1	13.577	13.577	1e-05	C	–	PRIMAL_DUAL_GAP_CERTIFIED
Extra restart	1000	267.96	176.38	0.342	U	–	MAX_ITER_REACHED

Clustering metrics below are percent, mean $\pm$ sample SD over the same fifty greedy K-means++/Lloyd seeds 0–49 on each saved G. Both metrics come from one complete endpoint. This is conditional

rounding variability, not independent datasets or full-pipeline uncertainty. A native-best assignment is not substituted for a mean, and its single-run value receives no across-run SD.

Table 7: Paired rounding protocol on the fixed current F/W. The IRLS300 flag refers only to completion of the declared historical-style 300-step protocol, including its iterate stopping rule; it does not certify optimality. Other stages are separate algorithms/restarts.

Stage	γ	ACC	NMI	n	IRLS300 complete	Evaluation status
SC on F	–	65.60 ± 1.87	80.31 ± 2.13	50	–	COMPLETED
IRLS300	1	65.56 ± 2.52	80.23 ± 2.34	50	no	COMPLETED
Uniform	1	65.40 ± 2.48	80.11 ± 2.28	50	–	COMPLETED
Diagonal	1	65.57 ± 2.36	80.08 ± 2.28	50	–	COMPLETED
Flow	1	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Warm restart	1	65.56 ± 2.52	80.23 ± 2.34	50	–	COMPLETED
IRLS300	1000	77.37 ± 4.49	84.84 ± 1.56	50	no	COMPLETED
Uniform	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Diagonal	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Flow	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Warm restart	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Extra restart	1	65.56 ± 2.52	80.23 ± 2.34	50	–	COMPLETED
Extra restart	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED

B.8 Fixed-representation graph evidence: G1–G3

This section uses the preregistered full ORL and cropped YaleB inputs, independently of retrospective subset winners. No best gamma, budget, graph or seed is selected here. Means and sample SD use the same fifty paired greedy K-means++/Lloyd seeds 0–49. A dash denotes unavailable evidence. C/U records certified/uncertified solver status; certification does not imply accurate clustering. The source files and hashes are listed in the generated report-input manifest.

The graph design contains 362 recorded comparisons across G1, G2 and G3. Scientific outcome and numerical qualification remain distinct.

Table 8: Graph-design coverage, including unsuccessful and unevaluated configurations. Shared mathematical problem identities are aliases, not independent measurements.

Phase	Observation	State	Tasks
G1	Scientific output	CASEBOOK_COMPLETE	2
G2	Scientific output	POINT_COMPLETE	48
G3	Scientific output	POINT_COMPLETE	312

G1: fixed gamma 1 casebooks. Casebook correctness uses an optimal assignment separately for each partition. Error-rate means below average the fifty paired rounding outcomes; deltas are SSC minus SC. They describe the fixed data and representation, not independent test datasets.

Table 9: Matched G1 means and sample SD at fixed gamma 1; these are not selected gamma-path maxima.

Data	SC ACC	SSC ACC	SC NMI	SSC NMI	SSC solver
ORL	78.33 ± 1.31	78.48 ± 1.41	89.87 ± 0.67	89.89 ± 0.80	PRIMAL_DUAL_GAP_CERTIFIED
YaleB cropped	21.78 ± 0.45	21.51 ± 0.48	27.73 ± 0.58	27.45 ± 0.58	PRIMAL_DUAL_GAP_CERTIFIED

Table 10: False merges and splits use contingency-count definitions. The averaging unit is a paired rounding seed, not a sample pair treated as an independent replicate.

Data	Pair error	SC (%)	SSC (%)	Delta (pp)	Pairs of rounds
ORL	False merge	0.91	0.92	0.01	50
ORL	False split	24.37	24.21	-0.16	50
YaleB cropped	False merge	7.78	7.95	0.17	50
YaleB cropped	False split	84.31	84.34	0.02	50

Table 11: Upstream A and shrinkage W0 contamination remain distinct. An unavailable A field is not replaced with W0 evidence. Fragmentation is class-induced and includes isolates.

Data	Graph / length space	Edges	Cross edges (%)	Cross mass (%)	Excess fragments
ORL	A: input-feature geometry	1273	22.31	11.99	8
ORL	W0: spectral geometry	1217	17.01	4.65	16
YaleB cropped	A: input-feature geometry	8510	58.88	53.60	438
YaleB cropped	W0: spectral geometry	7957	63.24	70.23	1109

Table 12: Predefined row-norm and weighted-degree quintiles, with original cut points and tie handling. Fractions pool the recorded sample-by-round outcomes within each fixed stratum.

Data	Variable	Quintile	n	Wrong degree	Wrong mass	SC right/SSC wrong (%)	SC wrong/SSC right (%)
ORL	row_norm	0	80	2.60	0.073	8.70	10.53
ORL	row_norm	1	80	1.06	0.118	5.78	6.17
ORL	row_norm	2	80	0.62	0.123	3.77	3.17
ORL	row_norm	3	80	0.36	0.160	4.23	4.03
ORL	row_norm	4	80	0.53	0.041	9.12	8.48
ORL	weighted_degree	0	80	2.33	0.021	10.25	10.57
ORL	weighted_degree	1	80	1.52	0.112	7.58	9.62
ORL	weighted_degree	2	80	0.60	0.116	5.88	5.58
ORL	weighted_degree	3	80	0.30	0.056	4.15	2.85
ORL	weighted_degree	4	80	0.42	0.211	3.75	3.75
YaleB cropped	row_norm	0	483	5.48	2.305	1.39	1.67
YaleB cropped	row_norm	1	483	5.19	2.347	3.14	2.79
YaleB cropped	row_norm	2	482	4.91	2.002	5.10	5.59
YaleB cropped	row_norm	3	483	3.83	1.170	7.89	6.15
YaleB cropped	row_norm	4	483	1.43	0.363	2.39	2.36
YaleB cropped	weighted_degree	0	483	2.58	0.157	4.94	4.85
YaleB cropped	weighted_degree	1	483	3.13	0.618	4.89	4.19
YaleB cropped	weighted_degree	2	482	4.09	1.255	4.23	4.07
YaleB cropped	weighted_degree	3	483	4.62	2.070	3.40	3.42
YaleB cropped	weighted_degree	4	483	6.42	4.087	2.45	2.02

Table 13: Deterministic first original row in each fixed seed-zero correctness category. Missing categories stay unavailable; no private image is embedded.

Data	Seed-zero category	Original row ID	Seed
ORL	sc_correct_ssc_wrong	1	0
ORL	sc_wrong_ssc_correct	271	0
ORL	both_correct	10	0
ORL	both_wrong	0	0
YaleB cropped	sc_correct_ssc_wrong	145	0
YaleB cropped	sc_wrong_ssc_correct	174	0
YaleB cropped	both_correct	0	0
YaleB cropped	both_wrong	2	0

G2: all 48 fixed cells. F0 is the saved raw spectral anchor; U is its row-normalized version. W0 and Wa are constructed in those respective geometries. Every graph is mass-matched to W0; equal mass does not equalize node forces or movement. Gamma zero is the anchor-only SC endpoint. Cached gamma-zero graph-arm rows are declared aliases, not independent fits.

Table 14: Complete fixed G2 path, including unavailable/failed cells. Movement is sum of row displacements divided by sum of anchor row norms; it is not Frobenius movement.

Data	Anchor	Graph	γ	ACC	NMI	Cert.	L21 move	Scientific outcome
ORL	F0	W0	0	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	W0	0.001	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	W0	1	78.48 ± 1.41	89.89 ± 0.80	C	0.0896	POINT_COMPLETE
ORL	F0	W0	10	77.68 ± 1.43	90.44 ± 0.56	C	0.15	POINT_COMPLETE
ORL	F0	W0	1000	74.52 ± 0.25	89.66 ± 0.18	U	0.302	POINT_COMPLETE
ORL	F0	W0	1e+06	38.25 ± 0.00	66.35 ± 0.00	U	0.678	POINT_COMPLETE
ORL	F0	Wa	0	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	Wa	0.001	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	Wa	1	77.83 ± 1.29	89.33 ± 0.69	C	0.0819	POINT_COMPLETE
ORL	F0	Wa	10	77.88 ± 1.21	89.49 ± 0.67	C	0.105	POINT_COMPLETE
ORL	F0	Wa	1000	78.07 ± 1.14	89.58 ± 0.60	C	0.132	POINT_COMPLETE
ORL	F0	Wa	1e+06	77.74 ± 1.32	89.59 ± 0.60	U	0.148	POINT_COMPLETE
ORL	U	W0	0	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	W0	0.001	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	W0	1	80.01 ± 1.54	90.94 ± 0.58	C	0.0452	POINT_COMPLETE
ORL	U	W0	10	79.92 ± 1.49	90.95 ± 0.40	C	0.0882	POINT_COMPLETE
ORL	U	W0	1000	74.82 ± 0.24	89.85 ± 0.09	U	0.257	POINT_COMPLETE
ORL	U	W0	1e+06	72.78 ± 0.76	89.35 ± 0.15	U	0.537	POINT_COMPLETE
ORL	U	Wa	0	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	Wa	0.001	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	Wa	1	80.34 ± 1.45	90.97 ± 0.53	C	0.0213	POINT_COMPLETE
ORL	U	Wa	10	80.33 ± 1.44	90.94 ± 0.57	C	0.0374	POINT_COMPLETE
ORL	U	Wa	1000	80.01 ± 1.51	90.89 ± 0.58	C	0.0578	POINT_COMPLETE
ORL	U	Wa	1e+06	80.14 ± 1.42	90.88 ± 0.61	U	0.0718	POINT_COMPLETE
YaleB cropped	F0	W0	0	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	W0	0.001	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	W0	1	21.51 ± 0.48	27.45 ± 0.58	C	0.0976	POINT_COMPLETE
YaleB cropped	F0	W0	10	21.60 ± 0.41	27.97 ± 0.33	U	0.335	POINT_COMPLETE
YaleB cropped	F0	W0	1000	16.88 ± 0.17	24.76 ± 0.61	U	0.714	POINT_COMPLETE
YaleB cropped	F0	W0	1e+06	9.82 ± 0.00	21.40 ± 0.00	U	0.838	POINT_COMPLETE
YaleB cropped	F0	Wa	0	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	Wa	0.001	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	Wa	1	22.53 ± 0.42	28.08 ± 0.53	C	0.16	POINT_COMPLETE

Data	Anchor	Graph	γ	ACC	NMI	Cert.	L21 move	Scientific outcome
YaleB cropped	F0	Wa	10	23.37 ± 0.45	28.93 ± 0.60	U	0.274	POINT_COMPLETE
YaleB cropped	F0	Wa	1000	3.40 ± 0.00	8.67 ± 0.00	U	0.964	POINT_COMPLETE
YaleB cropped	F0	Wa	1e+06	3.40 ± 0.00	8.67 ± 0.00	U	0.964	POINT_COMPLETE
YaleB cropped	U	W0	0	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	W0	0.001	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	W0	1	25.17 ± 0.85	31.88 ± 0.46	C	0.117	POINT_COMPLETE
YaleB cropped	U	W0	10	24.56 ± 0.50	30.91 ± 0.28	C	0.393	POINT_COMPLETE
YaleB cropped	U	W0	1000	15.08 ± 0.00	30.02 ± 0.00	U	0.82	POINT_COMPLETE
YaleB cropped	U	W0	1e+06	9.82 ± 0.00	21.40 ± 0.00	U	0.897	POINT_COMPLETE
YaleB cropped	U	Wa	0	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	Wa	0.001	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	Wa	1	26.11 ± 0.77	32.68 ± 0.53	C	0.0494	POINT_COMPLETE
YaleB cropped	U	Wa	10	26.49 ± 0.92	32.81 ± 0.62	C	0.165	POINT_COMPLETE
YaleB cropped	U	Wa	1000	28.30 ± 0.50	34.71 ± 0.32	U	0.447	POINT_COMPLETE
YaleB cropped	U	Wa	1e+06	3.40 ± 0.00	8.67 ± 0.00	U	0.957	POINT_COMPLETE

G3: fixed primary budget and gamma. The following rows use only the 15% addition budget and effective E8 gamma 1000. Additions come from the already frozen label-blind pool, with at most two newly incident edges per node and median positive W0 edge weight before mass matching. Oracle additions join same-class components after cross-class deletion and are reused unchanged in add-only and both arms. Other gammas and the 5% budget remain sensitivity records in the full CSVs, not alternative promotion gates.

Table 15: Fixed primary G3 arms. Blind controls retain exactly the Oracle deletion mask and randomize only additions. Same-class random additions do not prefer separate components. Fully randomized deletion/addition is a separate supplementary control. Infeasible or unrun controls retain their status.

Data	Arm / control	ACC	NMI	Cert.	L21 move	Excess fragments	Outcome
ORL	No edits	74.52 ± 0.25	89.66 ± 0.18	U	0.302	16	POINT_COMPLETE
ORL	Oracle delete	82.92 ± 1.22	92.95 ± 0.37	C	0.183	16	POINT_COMPLETE
ORL	Oracle add	68.57 ± 0.21	90.40 ± 0.04	U	0.457	1	POINT_COMPLETE
ORL	Oracle delete+completion	93.34 ± 1.15	97.98 ± 0.33	C	0.302	1	POINT_COMPLETE
ORL	Fixed delete + blind add 400	60.92 ± 0.22	86.70 ± 0.18	C	0.4	16	POINT_COMPLETE
ORL	Fixed delete + blind add 401	73.80 ± 0.67	91.57 ± 0.28	C	0.355	15	POINT_COMPLETE
ORL	Fixed delete + blind add 402	70.71 ± 0.17	89.79 ± 0.18	C	0.39	15	POINT_COMPLETE
ORL	Fixed delete + same-class random 400	87.19 ± 1.24	95.28 ± 0.37	C	0.244	11	POINT_COMPLETE

Data	Arm / control	ACC	NMI	Cert.	L21 move	Excess fragments	Outcome
ORL	Fixed delete + same-class random 401	85.97 ± 1.00	94.44 ± 0.24	C	0.226	13	POINT_ COMPLETE
ORL	Fixed delete + same-class random 402	86.24 ± 1.34	94.90 ± 0.31	C	0.228	13	POINT_ COMPLETE
ORL	Fully random delete/add 400	57.62 ± 0.62	83.55 ± 0.35	U	0.553	15	POINT_ COMPLETE
ORL	Fully random delete/add 401	68.11 ± 0.44	87.83 ± 0.19	U	0.516	19	POINT_ COMPLETE
ORL	Fully random delete/add 402	61.64 ± 0.96	84.51 ± 0.40	U	0.568	17	POINT_ COMPLETE
YaleB cropped	No edits	16.88 ± 0.17	24.76 ± 0.61	U	0.714	1109	POINT_ COMPLETE
YaleB cropped	Oracle delete	23.17 ± 0.64	30.63 ± 0.71	U	0.173	1109	POINT_ COMPLETE
YaleB cropped	Oracle add	11.64 ± 0.49	17.43 ± 0.83	C	0.869	53	POINT_ COMPLETE
YaleB cropped	Oracle delete+completion	54.04 ± 1.85	80.29 ± 0.84	U	0.764	53	POINT_ COMPLETE
YaleB cropped	Fixed delete + blind add 400	6.04 ± 0.07	11.96 ± 0.20	U	0.762	1000	POINT_ COMPLETE
YaleB cropped	Fixed delete + blind add 401	6.03 ± 0.10	11.63 ± 0.21	U	0.76	1023	POINT_ COMPLETE
YaleB cropped	Fixed delete + blind add 402	5.81 ± 0.10	11.58 ± 0.19	U	0.752	1016	POINT_ COMPLETE
YaleB cropped	Fixed delete + same-class random 400	41.48 ± 1.75	51.96 ± 0.93	C	0.532	661	POINT_ COMPLETE
YaleB cropped	Fixed delete + same-class random 401	42.25 ± 1.71	53.22 ± 0.79	C	0.538	640	POINT_ COMPLETE
YaleB cropped	Fixed delete + same-class random 402	40.96 ± 1.47	51.36 ± 0.71	C	0.517	684	POINT_ COMPLETE
YaleB cropped	Fully random delete/add 400	4.20 ± 0.05	9.70 ± 0.11	U	0.936	1376	POINT_ COMPLETE
YaleB cropped	Fully random delete/add 401	4.20 ± 0.05	10.02 ± 0.09	U	0.936	1368	POINT_ COMPLETE
YaleB cropped	Fully random delete/add 402	4.60 ± 0.06	10.57 ± 0.13	U	0.933	1380	POINT_ COMPLETE

Table 16: Preregistered promotion checks. Certification and activity are required for delete and both; blind-control certificate flags remain visible but are not an additional hidden pass condition. At least one development dataset must pass.

Data	Locked gate condition	Threshold	Satisfied	Decision
ORL	Delete and both certified	gap <= 1e-5; capacities <= 1e-8	yes	PASSED
ORL	Delete and both active	normalized L21 > 1e-6	yes	PASSED
ORL	Both minus delete ACC	mean gain >= 1 pp	yes	PASSED
ORL	Both minus delete NMI	mean gain >= 0 pp	yes	PASSED
ORL	Class-induced excess fragments	reduction >= 20 percent	yes	PASSED
ORL	ACC CI versus delete	lower endpoint > 0 pp	yes	PASSED
ORL	ACC CI versus mean blind additions	lower endpoint > 0 pp	yes	PASSED
ORL	Recorded excess-fragment reduction	93.75 percent	–	PASSED
YaleB cropped	Delete and both certified	gap <= 1e-5; capacities <= 1e-8	no	NOT_ EVALUABLE_ UNCERTIFIED
YaleB cropped	Delete and both active	normalized L21 > 1e-6	yes	NOT_ EVALUABLE_ UNCERTIFIED
YaleB cropped	Both minus delete ACC	mean gain >= 1 pp	yes	NOT_ EVALUABLE_ UNCERTIFIED

Data	Locked gate condition	Threshold	Satisfied	Decision
YaleB cropped	Both minus delete NMI	mean gain >= 0 pp	yes	NOT_ EVALUABLE_ UNCERTIFIED
YaleB cropped	Class-induced excess fragments	reduction >= 20 percent	yes	NOT_ EVALUABLE_ UNCERTIFIED
YaleB cropped	ACC CI versus delete	lower endpoint > 0 pp	yes	NOT_ EVALUABLE_ UNCERTIFIED
YaleB cropped	ACC CI versus mean blind additions	lower endpoint > 0 pp	yes	NOT_ EVALUABLE_ UNCERTIFIED
YaleB cropped	Recorded excess-fragment reduction	95.22 percent	–	NOT_ EVALUABLE_ UNCERTIFIED

Table 17: Stored paired percentile intervals: 10,000 resamples, bootstrap seed 424242. The three random-control outcomes are averaged within each rounding seed before the contrast. Only ACC intervals versus deletion and the mean blind control enter the gate; supplementary controls and NMI intervals are reported without redefining it.

Data	Fixed contrast	Metric	Mean delta (pp)	95% lower	95% upper
ORL	Both minus delete	ACC	10.41	9.92	10.91
ORL	Both minus delete	NMI	5.02	4.87	5.18
ORL	Both minus fixed-delete / blind-add mean	ACC	24.86	24.56	25.17
ORL	Both minus fixed-delete / blind-add mean	NMI	8.62	8.53	8.71
ORL	Both minus fixed-delete / same-class-random mean	ACC	6.87	6.53	7.23
ORL	Both minus fixed-delete / same-class-random mean	NMI	3.10	3.00	3.21
ORL	Both minus fully random edits mean	ACC	30.88	30.58	31.19
ORL	Both minus fully random edits mean	NMI	12.68	12.57	12.79
YaleB cropped	Both minus delete	ACC	30.87	30.39	31.34
YaleB cropped	Both minus delete	NMI	49.66	49.34	49.97
YaleB cropped	Both minus fixed-delete / blind-add mean	ACC	48.07	47.57	48.57
YaleB cropped	Both minus fixed-delete / blind-add mean	NMI	68.57	68.34	68.81
YaleB cropped	Both minus fixed-delete / same-class-random mean	ACC	12.47	11.92	13.04
YaleB cropped	Both minus fixed-delete / same-class-random mean	NMI	28.11	27.87	28.36
YaleB cropped	Both minus fully random edits mean	ACC	49.70	49.19	50.20
YaleB cropped	Both minus fully random edits mean	NMI	70.20	69.96	70.44

B.9 Conditional EnSC adaptation: capability and locked operation

The primary adaptation is the fixed $\tau = .95$ EnSC graph used only as the fusion graph for anchor F0. The source coefficient parameter is 50; graphs are matched to W0 total mass. Capability rows are label-assisted oracles: one complete configuration is selected by mean ACC, then NMI, then stable ID. The primary graph has five declared SSC gamma values; original SSC receives the frozen 15-value path. Controls remain separate named families. The locked operating row uses source $50/\tau = .95$/SSC gamma 1 without choosing parameters from labels. Both kinds of row use previously examined benchmarks; neither is an unseen-data test.

Table 18: Conditional G4/G5 results, percent mean plus sample SD over 50 matched greedy one-start roundings. Missing rows stay unavailable. C/U describe certified/uncertified fusion solves; native spectral controls have no SSC gap certificate. Search completeness and comparison validity are separate from completion of an individual selected row.

Data	Method / track	γ	ACC	NMI	Cert. / solver
ORL	EnSC $\tau = .95$ + SSC (oracle)	1	78.98 ± 1.26	91.40 ± 0.43	C / PRIMAL_DUAL_GAP_CERTIFIED
ORL	EnSC $\tau = .95$ + SSC (locked)	1	78.98 ± 1.26	91.40 ± 0.43	C / PRIMAL_DUAL_GAP_CERTIFIED
ORL	Fair SC: U (oracle)	0	80.17 ± 1.55	90.89 ± 0.64	– / ANALYTIC_G_EQUALS_F
ORL	Original SSC (oracle)	1	78.48 ± 1.41	89.89 ± 0.80	C / PRIMAL_DUAL_GAP_CERTIFIED
ORL	Angular SSC (oracle)	0.001	78.33 ± 1.31	89.87 ± 0.67	C / ANALYTIC_G_EQUALS_F
ORL	Sparse $\tau = 1$ + SSC (oracle)	1	79.16 ± 1.17	91.32 ± 0.52	C / PRIMAL_DUAL_GAP_CERTIFIED
ORL	Wider-kNN SSC (oracle)	0.001	78.33 ± 1.31	89.87 ± 0.67	C / ANALYTIC_G_EQUALS_F
ORL	Matched random SSC (oracle)	0.001	78.33 ± 1.31	89.87 ± 0.67	C / ANALYTIC_G_EQUALS_F
ORL	Native EnSC $\tau = .95$ (oracle)	–	81.68 ± 1.65	91.68 ± 0.65	– / NATIVE_SPECTRAL_CONTROL
ORL	Native EnSC $\tau = 1$ (oracle)	–	81.62 ± 2.08	91.55 ± 0.72	– / NATIVE_SPECTRAL_CONTROL
JAFFE	EnSC $\tau = .95$ + SSC (oracle)	1	96.41 ± 0.36	96.08 ± 0.54	C / PRIMAL_DUAL_GAP_CERTIFIED
JAFFE	EnSC $\tau = .95$ + SSC (locked)	1	96.41 ± 0.36	96.08 ± 0.54	C / PRIMAL_DUAL_GAP_CERTIFIED
JAFFE	Fair SC: U (oracle)	0	96.71 ± 0.00	96.39 ± 0.00	– / ANALYTIC_G_EQUALS_F
JAFFE	Original SSC (oracle)	10	95.31 ± 0.00	94.14 ± 0.02	C / PRIMAL_DUAL_GAP_CERTIFIED
JAFFE	Angular SSC (oracle)	1e+06	96.24 ± 0.00	95.52 ± 0.00	U / MAX_ITER_REACHED
JAFFE	Sparse $\tau = 1$ + SSC (oracle)	1	96.36 ± 0.31	95.99 ± 0.46	C / PRIMAL_DUAL_GAP_CERTIFIED
JAFFE	Wider-kNN SSC (oracle)	1000	95.31 ± 0.00	94.16 ± 0.00	C / PRIMAL_DUAL_GAP_CERTIFIED
JAFFE	Matched random SSC (oracle)	0.001	95.31 ± 0.00	94.13 ± 0.02	C / ANALYTIC_G_EQUALS_F
JAFFE	Native EnSC $\tau = .95$ (oracle)	–	99.06 ± 0.00	98.34 ± 0.00	– / NATIVE_SPECTRAL_CONTROL
JAFFE	Native EnSC $\tau = 1$ (oracle)	–	98.59 ± 0.00	97.78 ± 0.00	– / NATIVE_SPECTRAL_CONTROL
YaleB	EnSC $\tau = .95$ + SSC (oracle)	1	27.66 ± 0.74	36.99 ± 0.56	C / PRIMAL_DUAL_GAP_CERTIFIED
YaleB	EnSC $\tau = .95$ + SSC (locked)	1	27.66 ± 0.74	36.99 ± 0.56	C / PRIMAL_DUAL_GAP_CERTIFIED
YaleB	Fair SC: U (oracle)	0	25.80 ± 0.71	32.59 ± 0.43	– / ANALYTIC_G_EQUALS_F
YaleB	Original SSC (oracle)	0.001	21.78 ± 0.45	27.73 ± 0.58	C / ANALYTIC_G_EQUALS_F
YaleB	Angular SSC (oracle)	10	23.37 ± 0.45	28.93 ± 0.60	U / MAX_ITER_REACHED
YaleB	Sparse $\tau = 1$ + SSC (oracle)	1	27.59 ± 0.80	37.01 ± 0.51	C / PRIMAL_DUAL_GAP_CERTIFIED
YaleB	Wider-kNN SSC (oracle)	0.001	21.78 ± 0.45	27.73 ± 0.58	C / ANALYTIC_G_EQUALS_F
YaleB	Matched random SSC (oracle)	0.001	21.78 ± 0.45	27.73 ± 0.58	C / ANALYTIC_G_EQUALS_F
YaleB	Native EnSC $\tau = .95$ (oracle)	–	61.09 ± 1.17	66.49 ± 0.91	– / NATIVE_SPECTRAL_CONTROL
YaleB	Native EnSC $\tau = 1$ (oracle)	–	60.48 ± 1.37	66.04 ± 0.95	– / NATIVE_SPECTRAL_CONTROL
USPS	EnSC $\tau = .95$ + SSC (oracle)	0.001	65.60 ± 1.87	80.31 ± 2.13	C / ANALYTIC_G_EQUALS_F

Data	Method / track	γ	ACC	NMI	Cert. / solver
USPS	EnSC $\tau = .95$ + SSC (locked)	1	52.27 ± 3.95	69.73 ± 1.55	U / MAX_ITER_REACHED
USPS	Fair SC: U (oracle)	0	68.36 ± 3.64	83.44 ± 0.89	– / ANALYTIC_G_EQUALS_F
USPS	Original SSC (oracle)	30	66.53 ± 2.39	81.33 ± 1.20	U / MAX_ITER_REACHED
USPS	Angular SSC (oracle)	10	66.97 ± 3.23	81.95 ± 0.85	U / MAX_ITER_REACHED
USPS	Sparse $\tau = 1$ + SSC (oracle)	0.001	65.60 ± 1.87	80.31 ± 2.13	C / ANALYTIC_G_EQUALS_F
USPS	Wider-kNN SSC (oracle)	10	67.11 ± 2.51	81.87 ± 0.83	U / MAX_ITER_REACHED
USPS	Matched random SSC (oracle)	0.001	65.60 ± 1.87	80.31 ± 2.13	C / ANALYTIC_G_EQUALS_F
USPS	Native EnSC $\tau = .95$ (oracle)	–	49.51 ± 2.22	60.82 ± 1.19	– / NATIVE_SPECTRAL_CONTROL
USPS	Native EnSC $\tau = 1$ (oracle)	–	49.32 ± 2.00	60.73 ± 1.08	– / NATIVE_SPECTRAL_CONTROL

The native EnSC controls cluster each learned affinity in its own spectral space, with an explicit seeded eigensolver and the same fifty greedy one-start rounds. This differs from the source constructor default of 20 K-means starts. Gains over original SC/SSC must not be credited to SSC shrinkage when the native affinity clustering already explains them. The following differences compare the selected primary $\tau = .95$ + SSC row against matched native $\tau = .95$ EnSC, when both complete rows exist. They are descriptive differences after capability selection, not unbiased intervals or matched-movement evidence.

Data	ΔACC vs native EnSC (pp)	ΔNMI vs native EnSC (pp)
ORL	-2.70	-0.29
JAFFE	-2.65	-2.26
YaleB	-33.43	-29.50
USPS	16.09	19.49

Separate capability and locked-operation targets. Each formal row requires all four prescribed, qualified datasets with equal weights. ACC is compared with the stronger fair-SC/original-SSC baseline on each dataset; the two NMI contrasts retain those same baselines separately. Capability uses label-assisted parameter selection. Locked operation fixes source coefficient 50, $\tau = .95$ and SSC $\gamma = 1$ without choosing those parameters from labels. Both tracks are evaluated against the same label-assisted baseline selections, on previously examined benchmarks. Neither success criterion substitutes for the other.

Capability conclusion status: INCOMPLETE_FOUR_BENCHMARK_COMPARISON. Its formal macro and practical-target decision remain NA; partial datasets or unresolved certification/control coverage are not averaged into a result.

Locked-operation conclusion status: INCOMPLETE_FOUR_BENCHMARK_COMPARISON. Its formal macro and practical-target decision remain NA; partial datasets or unresolved certification/control coverage are not averaged into a result.

Table 20: Four-benchmark targets evaluated separately, in percentage points. Each target requires average ACC gain at least 2 pp and no average NMI loss against either fixed comparator. NA remains unavailable, not a zero or a negative result.

Track	Qualification	ΔACC strong	ΔNMI SC	ΔNMI SSC	Target met
Oracle capability	Unavailable	NA	NA	NA	NA
Locked $\gamma = 1$	Unavailable	NA	NA	NA	NA

The parameter-selection-gap flag is unavailable because both formal, qualified four-benchmark target decisions are required. No parameter-selection diagnosis is inferred from a partial track.

Table 21: Per-dataset selected-capability contrasts, in percentage points. Unqualified rows are diagnostic observations only and do not enter a formal macro conclusion. Negative values and losses remain visible.

Data	Complete/intended	ΔACC strong	ΔNMI SC	ΔNMI SSC	Qualification
ORL	44/44	-1.20	0.51	1.51	unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN
ORL_att92: observed ACC loss exceeds 1 percentage point against the stronger fair baseline; qualification above still applies.					
JAFFE	44/44	-0.30	-0.31	1.94	unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN
YaleB	44/44	1.86	4.39	9.26	unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN
USPS	44/44	-2.76	-3.12	-1.02	unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN
USPS: observed ACC loss exceeds 1 percentage point against the stronger fair baseline; qualification above still applies.					

B.10 Coverage, solver qualification and cost

Computational cost is unavailable.

Graph criterion. Oracle intervention criterion: PASSED. A formal four-benchmark performance conclusion requires complete, qualified controls on all four datasets.

Generated-input provenance. Tables and figures are generated from the authoritative committed CSV/JSON records only. The input hash manifest is `results/submission_refresh/report_input_manifest.json`. Generated snapshot date: 2026-11-10.

C Additional protocol and implementation detail

The source/subset catalogue fixes AR seven-of-fourteen rules, UMIST deduplication/omission hypotheses and BinAlpha 30-per-letter draws with explicit original indices. Diagnostic parents cannot compete as equivalent paper reconstructions. ORL/JAFFE membership is not arbitrarily searched. Yale source, resize interpolation and Y0–Y3 normalization factors remain separate; USPS uses all 9,298 rows in compatible experiments. Native 16-by-16 MSRA is labelled as an adapted-resolution source and is not upsampled to claim recovery of the paper's 32-by-32 data. A changed subset always rebuilds both graphs and both representations. Complete coverage, locked selections and confirmation records, rather than numerical closeness alone, determine the reported scope.

The instrumented ClustPy port pins commit `4ac9d00685efc7432800684f5bfac0f01af59172`[1]. It preserves failed-update continuation and both the actually used and source-returned projections. Scoped NumPy seeding of the source PCA is the disclosed determinism modification. The separately named PCA-preconditioned ridge LDA-Kmeans→SC adaptation uses centered float64 input, numerical rank tolerance $\epsilon \max(n,p)\sigma_{\max}$, $p_0 = \min(100, p, n-c, \mathrm{rank}(X_c))$, $r = \min(c-1, p_0)$ and ridge $\eta = 10^{-6}\mathrm{tr}(S_t)/p_0$. Covariance scatter uses divisor $n-1$; inner K-means has ten starts, the outer cap is 300, and stopping compares partitions up to label permutation. The actually clustered projection is returned. Raw SC, PCA_{p_0} SC and PCA_r SC controls use one RANDOM Lloyd start per matched external seed; direct learned-label versus downstream SC contrasts therefore also reflect their prescribed rounding budgets. Neither this adaptation nor a degenerate reference execution recovers historical LDA-SC.

The diagonal PDHG formulation uses $K = \mathrm{diag}(\sqrt{w})B$ and maps its dual back to original capacities $2\gamma w$; independent small convex references validate the target. Agreement between numerical solvers alone is not a certificate. Orthogonal eigenspace rotations preserve Euclidean distances, while a

[1] `https://github.com/collinleiber/ClustPy/blob/4ac9d00685efc7432800684f5bfac0f01af59172/clustpy/centroid/ldakmeans.py`; the original source, BSD-3-Clause notice and exact hashes are retained.

changed cutoff eigenspace or row normalization can change clustering. Edge deletion need not preserve $G = F$: on an eight-node cycle at $\gamma = 0.6$, removing one edge destroys force cancellation. The separately valid incident-mass sufficient bound is $2\gamma \max_i \sum_j w_{ij} \leq 1$.

G3 holds the full development F_0 fixed and uses a label-blind top-20-neighbor addition pool for oracle evaluation. The fixed 15%/gamma-1000 gate requires certified active deletion and deletion-plus-completion solves, at least one percentage point paired ACC gain, nondecreasing NMI, at least 20% fewer excess class fragments, and positive conditional bootstrap intervals against deletion and fixed oracle deletion plus mean blind random additions. Fully randomized deletion/addition and same-class random additions remain separate controls. The labelled intervention diagnoses available graph information; it is not a label-free algorithm or an unseen-data test. Subsequent EnSC capability and locked-operation targets require their own complete four-benchmark qualification; neither can be inferred from the gate alone.

The EnSC port targets the true elastic-net coefficient objective and is checked against independent small convex references. Its row-wise KKT diagnostics remain separate from SSC primal–dual certificates: a C/U fusion mark does not certify coefficient optimality. On USPS, 969 of 9,298 tau-0.95 coefficient rows fail the declared KKT diagnostic, despite source screening/objective stops. The fixed gamma-1 fusion gap is 2.8042×10^{-5}, above the certification threshold. Original-SSC gamma 30, its capability choice, is also uncertified. The USPS matched-movement comparison returns the degenerate zero-movement match $G = F$; ORL/JAFFE/YaleB have no certified movement bracket. These observations do not identify a topology-specific benefit at equal nonzero shrinkage.

D Nonconvex fusion: scope and negative evidence

Truncated fusion is established in PRclust [Pan et al., 2013, Wu et al., 2016]; distance-dependent robust pairwise fusion is central to Shah–Koltun RCC [Shah and Koltun, 2017]. These differ from Sun et al.'s resistant convex clustering [Sun et al., 2025]. The SSC adaptation does not claim invention of these penalties.

For two concentrated groups of sizes $m < n$, separation d and aggregate attraction λ, the unsquared-fidelity linear-fusion optimum is $d \min(m, \lambda)$. With capped penalty $\min(r, \tau)$ and $d > \tau$, it is $\min(md, \lambda\tau)$. Thus $m < \lambda < md/\tau$ gives fusion for the linear objective and separation for the capped objective. Equality can admit multiple minimizers. These two-representative statements are not arbitrary-graph clustering guarantees.

A separately enumerated five-point graph has linear optimum 8 and capped optimum 3. The local capped algorithm obtains 3 from F but 8 from the original SSC output: accurate inner convex solves do not certify a global nonconvex solution. Synthetic imbalanced-bridge experiments also show the limitation of damage avoidance: SSC averages 99.156 ACC/96.725 NMI while capped fusion, static screening and squared fidelity recover 100/100, which SC already achieves. Capping is therefore neither necessary in that example nor evidence of additional clustering capacity over SC. Historical synthetic and secondary geometry results remain distinct from the present graph-adaptation comparison.

The mathematical and empirical observations motivate a narrow conclusion: representation motion, robustness or objective improvement alone is insufficient. A useful extension must retain both ACC and NMI improvements over appropriate SC, SSC and affinity-only controls, with numerical qualifications attached to the exact saved outputs.